

INTERNAL CTO BRIEFING

Insight Projects

A secure, Arabic-first workspace from program plan to verified result.



إنسایت بروجکٹس

Product explanation, technical posture, likely CTO questions, test evidence, and pilot recommendation.

Recommended decision

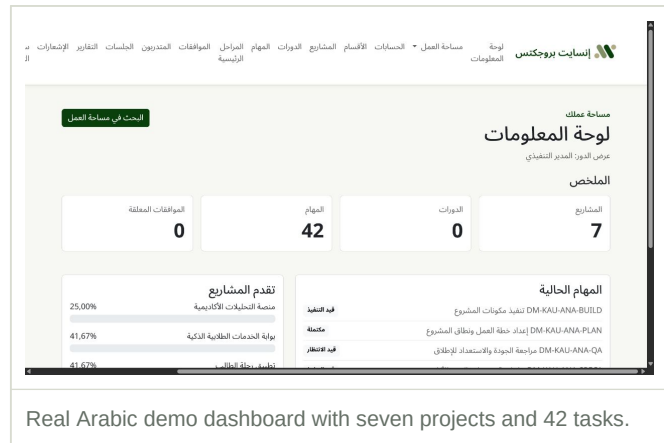
Approve a controlled pilot for one program using fictional or anonymized data. Production use remains gated by university security, data-residency, operational ownership, and restore-test approval.

EXECUTIVE OVERVIEW

What the product is - and why the university would use it

Insight Projects is a Django and PostgreSQL application that centralizes programs, projects, courses, tasks, people, attendance, approvals, evidence, notifications, dashboards, and exports in one permission-controlled workspace.

- Leadership sees current progress, overdue work, approval status, and operational evidence without waiting for manually assembled updates.
- Delivery teams work from one hierarchy with explicit ownership and a durable activity trail.
- Arabic is a first-class interface and reporting language; English is available with direction-aware RTL/LTR behavior.
- Important business records are archived instead of silently deleted.



Real Arabic demo dashboard with seven projects and 42 tasks.

One-sentence answer

"It is an internal delivery-control platform that connects the plan, the people, the approvals, and the evidence in Arabic and English."

FICTIONAL UNIVERSITY SCENARIO

How programs, projects, tasks, and shared workers fit together

Program	Example projects	People model
KAU digital services	Smart Student Services Portal; Academic Analytics Platform	KAU specialists plus a shared coordinator and shared analyst
KSU research enablement	Research Innovation Incubator; Laboratory Management Platform	KSU specialists plus the same cross-program resources
KKU student experience	Student Journey Application; Digital Support Center	KKU specialists plus shared roles where authorized

The demo data is fictional. It is designed to prove that the same task pattern can repeat across projects while project-specific work and membership remain separate.

إنساييت بروجكتس

لوحة معلومات

مساحة العمل

الحسابات

الأقسام

المشاريع

الدورات

المهام

المراحل الرئيسية

الموافقات

المتدربون

الجلسات

التقارير

الإشعارات

ال

إنشاء مشروع

التصنيفات

العملاء

المشاريع

المشاريع المتاحة وفقاً لصلاحيات دورك.

البحث في المشاريع

جميع الحالات

جميع الأولويات

تضمين المؤرشف

تصفية

المشروع	الحالة	الأولوية	مدير المشروع	التواريخ
DEMO-KAU-ANALYTICS — منصة التحليلات الأكاديمية	نشط	عالية	مدير برنامج جامعة الملك عبدالعزيز	2026-08-08 – 2027-03-06
DEMO-KAU-PORTAL — بوابة الخدمات الطلابية الذكية	نشط	حرجة	مدير برنامج جامعة الملك عبدالعزيز	2026-07-01 – 2027-03-01
DEMO-KKU-JOURNEY — تطبيق رحلة الطالب	نشط	عالية	مدير برنامج جامعة الملك خالد	2026-07-01 – 2027-03-21
DEMO-KKU-SUPPORT — مركز الدعم الرقمي	نشط	متوسطة	مدير برنامج جامعة الملك خالد	2026-08-28 – 2027-03-26
DEMO-KSU-INNOVATION — حاضنة الابتكار البحثي	نشط	عالية	مدير برنامج جامعة الملك سعود	2026-07-01 – 2027-03-11
DEMO-KSU-LABS — منصة إدارة المختبرات	نشط	متوسطة	مدير برنامج جامعة الملك سعود	2026-08-18 – 2027-03-16
KKU — جامعة الملك خالد	مسودة	عالية	user fiction	2026-07-07 – 2026-12-07

Real Arabic project register showing fictional KAU, KSU, and KKU projects.

What to demonstrate

Open the project register, select KAU's Smart Student Services Portal, point out the bilingual name, client, category, project manager, supervisor, priority, team, progress, budget, dates, and approval action.

OPERATING WORKFLOW

What each team member experiences

Step	Actor	System behavior
1. Plan	Executive / Project Manager	Create the project, team, tasks, milestones, dates, priority, and responsibilities.
2. Execute	Employee / Contractor	Update assigned work, progress, comments, tags, and permitted evidence.
3. Review	Supervisor	Review first-stage completion and approve or reject with a recorded reason.
4. Approve	Project Manager	Provide the second independent decision. The same person cannot approve both stages.
5. Govern	Executive / authorized roles	View dashboard, overdue work, audit records, and scoped PDF/XLSX reports.

Technical Administrator Manages technical configuration and internal accounts; not a shortcut around object permissions.	CEO / Executive Manager Portfolio-level visibility and governance according to granted scope.
Project Manager / Supervisor Own project delivery and perform distinct approval stages.	Employee / Contractor Work within assigned projects and tasks; direct URL checks still enforce access.
External Trainer Uses a random, expiring, session-scoped capability link for attendance only.	Concurrent work Multiple users can use separate devices at the same time; transactions and database constraints protect critical writes. Final capacity still needs load testing.

Approval rule to remember

Supervisor first, Project Manager second. One actor cannot complete both stages. Rejections require a reason, retries are preserved, and history is not rewritten.

PRODUCT CAPABILITY

What is already implemented

Program delivery

Projects, departments, clients, categories, courses, tasks, milestones, teams, progress, archives, Kanban, calendar, timeline, and Gantt views.

People and attendance

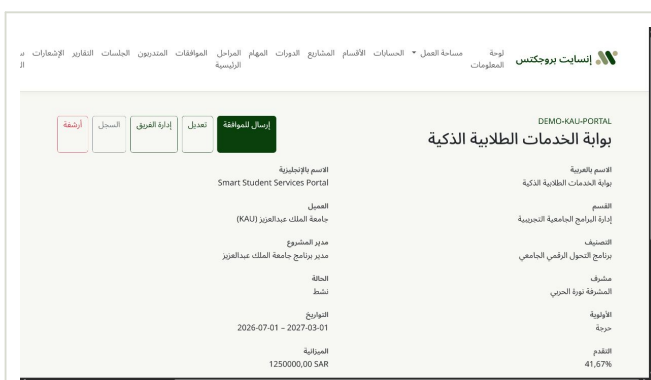
Internal users, trainees, imports, sessions, trainer links, submission, review, correction, and aggregate attendance reporting.

Governance

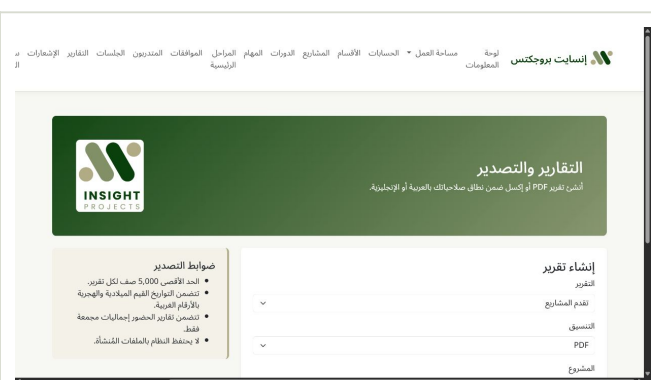
Sequential approvals, append-only audit history, scoped search, notifications, operational archive and restore flows.

Reports

Project progress, overdue tasks, and attendance summaries in PDF/XLSX; Arabic or English; Gregorian and Hijri dates with Western digits.



KAU project detail and team controls.



Arabic PDF/XLSX report center.

Export safeguards: at most 5,000 rows, at most a 366-day range, aggregate attendance only, permission-scoped rows, spreadsheet formula neutralization, no persistent generated-file URL, and no-store responses.

TECHNICAL ARCHITECTURE

A maintainable monolith with clear production boundaries

Layer	Technology / responsibility	Why it matters
Web application	Django 5.2 LTS, Python 3.13, Django Templates, Bootstrap 5, HTMX	Simple server-rendered architecture with fewer moving parts and first-class localization.
Data	PostgreSQL in development, test, staging, and production	Transactions, constraints, indexes, durable relational integrity, and scoped queries.
Background work	Celery 5 and Redis 8 when background processing is required	Separates notifications or longer work from interactive requests.
Runtime	Docker / Compose locally; Gunicorn for production	Repeatable packaging and a portable hosting target.
Files	Local private storage in development; S3-compatible private storage in production	Avoids public-by-default evidence files and supports controlled delivery.
Observability	Structured logs, request correlation, health checks, audit records	Supports diagnosis without exposing stack traces or raw sensitive metadata.

Deployment stance

The application is portable. Render is documented as a target, but the university must approve provider, region, plan, domain, network posture, and data residency.

Scalability answer

The web tier can be replicated and the database/background workers scaled, but university-scale concurrency has not yet been load-tested. A pilot supplies realistic usage assumptions.

Integration stance

The current release is a monolith, not an integration marketplace. SSO, institutional systems, and API integration should be explicitly scoped after the pilot.

Maintenance stance

Domain apps, thin views, service-layer workflows, migrations, typed interfaces, and automated tests keep change risk understandable.

Honest architecture answer

"The architecture is ready for a controlled pilot. Production topology and integrations are decisions to complete with university infrastructure and security teams, not assumptions hidden inside the demo."

SECURITY, PRIVACY, AND ARABIC

Controls already present - and controls still required before launch

Access control

Authenticated sessions, active-account checks, role plus object-level authorization, method checks, CSRF protection, and permission-scoped exports.

Sensitive workflows

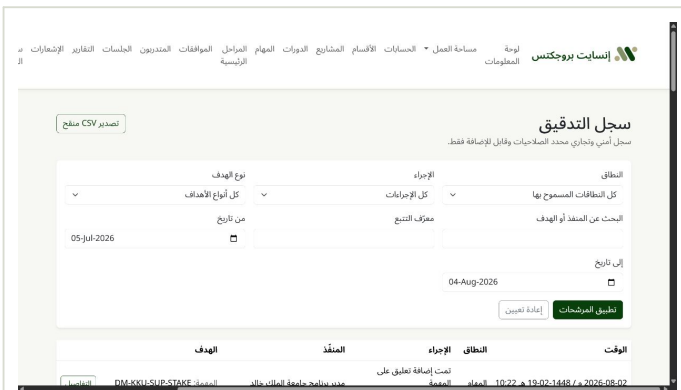
Database transactions for approvals/imports, immutable approval history, randomized expiring trainer links, hashed tokens where practical, and replay/expiry checks.

Data protection

Validated upload size/extension/MIME, private production object storage, no production DEBUG, secrets from environment variables, no raw sensitive audit export.

Arabic quality

Arabic is preserved, not destructively normalized. Search uses approved normalization. RTL/LTR, mixed content, Unicode round-trip, PDFs, XLSX, and Hijri/Gregorian rendering are verified.



Arabic append-only, permission-scoped audit view.

Operational targets

- Backup target: RPO 24 hours and RTO 8 hours.
- Encrypted backup retention target: 35 days.
- Quarterly isolated restore exercise is required.
- Application data retention is indefinite for the initial release unless policy changes.

Not yet claimed

- No real production backup has been created or restored.
- No production provider/region/data-residency approval yet.
- Named operational owners and alert routes remain a launch gate.

VERIFICATION EVIDENCE

What was tested, what passed, and what has not been claimed

Evidence	Latest verified result	Coverage
Automated non-browser	159 passed	Models, constraints, selectors, forms, permissions, services, transactions, reports, security cases, imports, and integrations.
Playwright browser	14 passed	Critical Arabic RTL and English LTR journeys, permissions, navigation, report workflows, and responsive/mobile behavior.
Total test suite	173 collected	Latest full local run on 2026-08-03; collection reconfirmed as 173 on 2026-08-04.
Static quality	Passed	Ruff formatting/checks; mypy across 171 source files; Django system checks; no migration drift.
Localization	Passed	1,007 Arabic messages compiled with 0 untranslated; Arabic round-trip, mixed content, RTL/LTR, Hijri/Gregorian output.
Manual artifact QA	Passed	PDF pages rendered and visually inspected; Excel output opened/inspected; Arabic and English reports checked.

Unit tests Pure business rules, formulas, validators, token utilities, invalid/boundary cases, archived state, and retry behavior.	Integration tests Django views/templates, PostgreSQL constraints and transactions, object permissions, imports, report policies, query-count and security boundaries.
Browser tests Real user journeys with Playwright across Arabic/English and representative responsive viewport behavior.	Security tests Anonymous/inactive/wrong-role/wrong-object access, direct URLs, CSRF/method restrictions, uploads, token expiry/replay, sensitive logs, and spreadsheet formula injection.

Do not overclaim

Formal penetration testing, university-scale load testing, disaster-recovery restoration, provider security review, dependency/vulnerability scans at release time, and production UAT remain required before live launch.

LIKELY CTO QUESTIONS

Concise answers you can use in the meeting

Why build this instead of using Jira, Microsoft Project, or Asana?

This is a domain-specific internal workflow that combines project delivery, courses, trainee attendance, two-stage approvals, Arabic reporting, and university-specific access in one controlled model. The pilot should prove where this consolidation adds value; it does not need to replace every specialist tool on day one.

Where will the data be hosted, and does it meet Saudi data-residency requirements?

The application is portable and containerized, but the final provider, Saudi region, network model, and data-residency position are not assumed. Those are explicit university decisions before real data is used.

Does it support our SSO or identity provider?

The current release uses secure internal Django accounts. Institutional SSO is not claimed as complete. We can scope SAML/OIDC integration with the university IdP after confirming its standards, role mapping, lifecycle rules, and MFA requirements.

Can many users work at the same time?

Yes, separate devices and sessions can work concurrently, and critical writes use PostgreSQL transactions and constraints. The precise safe capacity has not been claimed; a realistic load test is a pre-production gate.

How do you stop users from seeing another program?

Every protected view and export applies role and object-level scope. Direct URLs are checked, not just hidden in navigation. Tests cover anonymous, wrong-role, wrong-object, and cross-scope attempts.

Can records be deleted or the audit trail rewritten?

Important records use archive/restore workflows. Approval history is immutable and audit records are append-only by design. Exported audit data is deliberately sanitized.

How are backups and disaster recovery handled?

The documented target is encrypted backups with a 24-hour RPO, 8-hour RTO, 35-day backup retention, and quarterly isolated restore tests. A real provider backup and successful restore evidence are mandatory before launch.

Is Arabic truly supported or only translated labels?

Arabic is first-class: RTL layout, preserved Unicode text, approved search normalization, mixed Arabic/English behavior, Arabic PDF/XLSX, and Hijri plus Gregorian dates were tested. English remains available with LTR layout.

DECISION AND NEXT STEPS

How to answer remaining questions and close the meeting

Is it production-ready today?

Feature-complete and verified locally, yes. Approved for live university data, not yet. Production requires hosting/data-residency approval, SSO decision, secrets and storage configuration, vulnerability scanning, monitoring/alert ownership, real backup restoration, load testing, and role-based UAT.

What about mobile?

It is a responsive web application tested at representative mobile viewports. It is not a native iOS or Android application.

Can we integrate with university systems?

The architecture can support scoped integrations, but no specific SIS, HR, LMS, email, or SSO integration should be promised until interfaces, ownership, data classifications, and failure behavior are agreed.

Who will operate it after the pilot?

A production owner, infrastructure owner, security contact, support route, alert route, and change process must be named. The current repository includes deployment and operations runbooks, but people and accountability complete the control.

Pilot decision	Recommended scope	Success evidence
One program	Fictional or anonymized data; defined roles; 4-8 week time box; no production integration dependency	Users complete assigned workflows; approval trace is clear; leadership report is trusted; Arabic UAT issues are recorded and resolved
Technical gates	Approved hosting region; security review; backup and restore exercise; monitoring owner; dependency scan; load-test target	Signed gate checklist with evidence and named owners
Go / no-go review	CTO, security, operations, product owner, and Arabic business reviewer	A documented decision on production, remediation, or further pilot work

Close with this

"I am asking for permission to run a controlled pilot, not permission to bypass production governance. The product is ready to demonstrate; the pilot will generate the technical and operational evidence needed for a responsible production decision."

PRESENTER RUNBOOK

Your three-minute live demo and meeting checklist

Three-minute live demo sequence

- 0:00-0:30 - State the problem and the one-sentence value.
- 0:30-1:15 - Show the Arabic dashboard and the KAU/KSU/KKU project register.
- 1:15-2:00 - Open the KAU project, explain shared workers, tasks, and two-stage approvals.
- 2:00-2:35 - Show reports and the audit trail; mention Hijri/Gregorian dates and scoped exports.
- 2:35-3:00 - State the verified test evidence, the honest production gates, and ask for the pilot.

Before the meeting Start Docker and the application; sign in as the executive demo user; select Arabic; preload Dashboard, Projects, the KAU project, Reports, and Audit in separate tabs.	If the live demo fails Continue with the HTML deck screenshots. Do not troubleshoot infrastructure during the three-minute pitch; offer a technical follow-up session.
Avoid overpromising Say "implemented locally" for features and "required production gate" for SSO, hosting approval, load tests, penetration testing, restore evidence, and live-data approval.	Meeting outcome Ask for a named pilot sponsor, one program, fictional/anonymized data, 4-8 weeks, agreed success measures, and a technical/security review date.

Your strongest answer

"The product is ready for a controlled pilot, and the remaining risks are visible, testable, and assigned to explicit production gates."

Evidence and repository references

Repository sources: docs/SCOPE.md; BUSINESS_RULES.md; USER_ROLES.md; WORKFLOWS.md; ACCEPTANCE_CRITERIA.md; LOCALIZATION.md; SECURITY.md; TEST_STRATEGY.md; DEPLOYMENT.md; OPERATIONS.md; PHASE_13_VERIFICATION.md; PHASE_14_VERIFICATION.md.